Multi-Agent Reinforcement Learningfor No-Limit Texa

Aléo Ibrahim Houmed & Maxence Adly

Reinforcement Learning Group Project

1. Introduction & Motivation

No-Limit Texas Hold'em (NLHE) is one of the most challenging domains in AI due to its combination of:

- **Imperfect information**: players cannot see opponents' cards
- **Stochastic dynamics**: card dealing introduces randomness
- **Large action space**: bet sizing creates combinatorial complexity
- **Multi-agent competition**: opponents adapt their strategies
- **Sparse, delayed rewards**: outcomes revealed only at hand conclusion

We train autonomous agents to play NLHE through **self-play reinforcement learning**, where a single shared policy improves by competing against copies of itself. This approach has proven effective in games like Go (AlphaGo/Zero) and poker (Pluribus, DeepStack).

Key contributions:

- Custom multi-agent NLHE environment supporting 2–10 players
- PPO training with action masking for legal move enforcement
- Cash-game mode with persistent stacks and player elimination
- Interactive GUI and CLI tools for model evaluation

2. Game & Environment Design

Environment specifications:

Observation Space (per player):

- Hole cards (one-hot, 52-dim)
- Board cards (multi-hot, 52-dim)
- Street indicator (one-hot, 4-dim)
- Position relative to dealer
- Normalised stacks, pot, bets
- Min/max raise amounts
- Per-street betting history

All values normalised to $[0, 1]$.

Action Space (6 discrete actions):

Idx	Action
0	Fold
1	Check / Call
2	Raise Minimum
3	Raise $\frac{1}{2}$ Pot
4	Raise Pot
5	All-In

Reward: $r_i = \text{stack}_i^{\text{final}} - \text{stack}_i^{\text{initial}}$ Sparse, zero-sum at hand end.

3. Method: PPO with Self-Play

We use **Proximal Policy Optimization (PPO)** within the **Ray RLlib** framework, employing a single shared policy for all agents (*self-play*).

PPO Objective (clipped surrogate):

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min\Big(r_t(\theta)\, \hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\, \hat{A}_t \Big) \right]$$

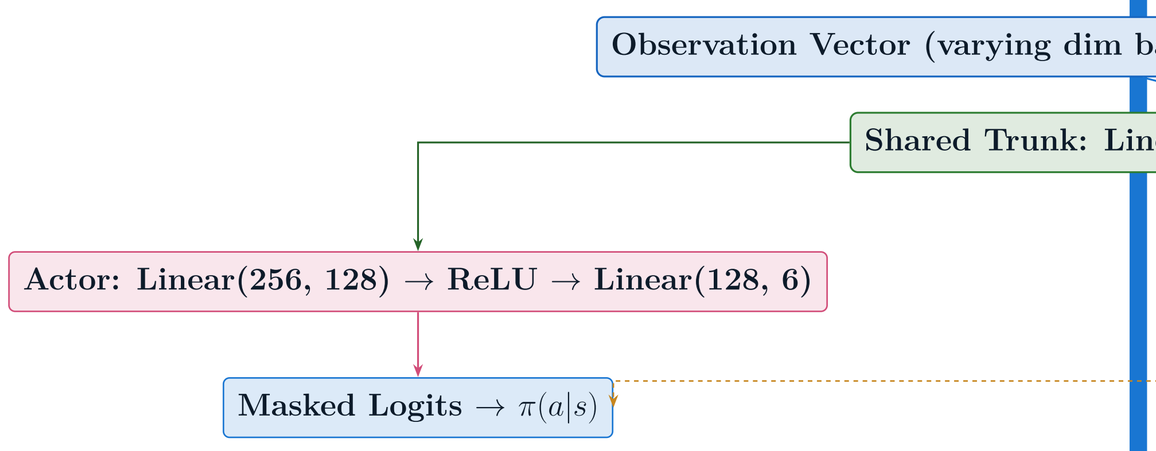
where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ and $\hat{A}_t$ is the GAE advantage estimate.

Action Masking: Illegal actions are suppressed by setting their logits to -10^{38} before the softmax, guaranteeing $\pi(a|s) = 0$ for illegal moves:

$$\ell'_i = \begin{cases} \ell_i & \text{if action } i \text{ is legal} \\ -10^{38} & \text{otherwise} \end{cases}$$

Hyperparameter	Value
Learning rate	3×10^{-4}
Discount γ	1.0 (episodic)

4. Neural Network Architecture



The **PokerActionMaskRLModule** extends RLlib's `DefaultPPOTorchRLModule`:

- **Shared trunk**: Extracts features from the observation vector
- **Actor head**: Produces 6-dimensional logits, masked for legality
- **Critic head**: Outputs scalar baseline value $V(s)$ for variance reduction
- Total parameters: ~200K (lightweight, fast inference)